

Auteur: Noran Ben Said
Studierichting: Informatica
Oefeningenreeks 4A nr. 56.3

$$I = \int \sqrt{1 - \sin(2x)} dx$$

$$1 = \sin^2 x + \cos^2 x$$

$$\sin(2x) = 2 \sin x \cos x$$

Dus:

$$1 - \sin(2x) = \sin^2 x + \cos^2 x - 2 \sin x \cos x = (\sin x - \cos x)^2$$

$$I = \int \sqrt{(\sin x - \cos x)^2} dx = \int |\sin x - \cos x| dx$$

Geval 1: $\sin x \geq \cos x$

$$I = \int (\sin x - \cos x) dx = -\cos x - \sin x + c$$

Geval 2: $\sin x \leq \cos x$

$$I = \int (\cos x - \sin x) dx = \sin x + \cos x + c$$

References